# 🗓️ Document 4: Product Delivery & Sprint Execution Plan

**Product Name:** ClinikFlow (India Specialist Edition)
**Document Version:** 1.0.0
**Methodology:** Agile Scrum (2-Week Iterations)
**Total Timeline:** 12 Weeks (6 Sprints) to Full Commercial Pilot

## 1. Master Release Roadmap

```
Week  1 - 2  : [ Sprint 1 ] — Core Token Engine & Compounder Desk
Week  3 - 4  : [ Sprint 2 ] — Doctor Mobile Controller & Waiting Room TV
Week  5 - 6  : [ Sprint 3 ] — WhatsApp Automation & Zero-App Patient Web Tracker
Week  7 - 8  : [ Sprint 4 ] — Specialist EMR Lite & Rapid Rx Engine
Week  9 - 10 : [ Sprint 5 ] — Dynamic UPI Payments, Thermal Slip & Daily Ledger
Week 11 - 12 : [ Sprint 6 ] — Offline-First PWA Sync, UAT & Live Clinic Pilot
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## 2. Sprint-by-Sprint Work Breakdown Structure (WBS)

### Sprint 1: High-Speed Token Desk & State Core (Weeks 1–2)

*Primary Goal:* Receptionist can register walk-ins and issue tokens in under 12 seconds with local persistence.

- **User Stories & Engineering Tasks:**
  - `TASK-101`: Establish database schemas (Patients, Clinics, DailyQueueTokens) using PostgreSQL + Prisma/Drizzle.
  - `TASK-102`: Build the **10-Second Compounder Desk**:
    - 10-digit mobile number lookup with instant debounce match.
    - Keyboard shortcuts (`Alt+N` new token, `Enter` to confirm, auto-focus).
    - Auto-assignment of sequential daily token number.

    - `TASK-103`: Implement `BroadcastChannel` local event bus for zero-latency multi-tab updates on the reception machine.
    - `TASK-104`: Unit tests for token state machine (`WAITING`, `CALLED`, `COMPLETED`, `CANCELLED`).
- **Milestone Gate 1:** Compounder can register 25 simulated patients in under 5 minutes without touching a mouse.

## Sprint 2: Doctor Mobile Controller & TV Display (Weeks 3–4)

*Primary Goal:* Doctor controls queue flow from phone/tablet; TV display updates instantly with audio.

- **User Stories & Engineering Tasks:**
    - `TASK-201`: Build responsive **Doctor Mobile Cabin Controller**:
        - Sticky top ribbon with `[Call Next]`, `[Hold for Lab]`, `[Complete]`.
        - Up-next patient drawer showing age, gender, and chief complaints.
    - `TASK-202`: Build **Waiting Room Smart TV Kiosk View (`tv-display.html`)**:
        - High-contrast layout readable from 25 feet.
        - Smooth CSS entrance transitions for "Now Serving".
    - `TASK-203`: Implement **Web Audio API + Vernacular Synthesizer**:
        - Pleasant dual-tone hospital chime ($E_5 \rightarrow C_5$ E5 → C5).
        - Spoken voice announcement in Hindi and English ("*टोकन नंबर 14, कृपया डॉक्टर केबिन में आएं*").
- **Milestone Gate 2:** Doctor tapping "Next" on an iPhone triggers chime and animates the TV display within 250 milliseconds.

## Sprint 3: WhatsApp Automation & Patient Web Tracker (Weeks 5–6)

*Primary Goal:* Patients track their live queue position without downloading any application.

- **User Stories & Engineering Tasks:**
    - `TASK-301`: Integrate WhatsApp Business Cloud API:
        - Approved template for instant token confirmation.
        - Dynamic deep link generation (`clinik.in/q/{token_uuid}`).
    - `TASK-302`: Build lightweight **Patient Live Tracker Web App (PWA)**:
        - Total bundle size < 150KB.

        - Displays: Current token serving, tokens ahead, estimated wait time.
        - Dynamic color change (Blue when far → → Green when top 3 → → Gold when called).
    - `TASK-303`: Implement the **"Report Check" (Re-entry) Queue Workflow**:
        - Doctor taps `[Hold for Diagnostics]`.
        - Compounder marks `[Reports Arrived]`.
        - System interleaves patient into next available priority slot.
- **Milestone Gate 3:** Real patient receives WhatsApp alert on their phone and tracks queue movement in real time over 4G.

## Sprint 4: Specialist Clinical Module & Rx Engine (Weeks 7–8)

*Primary Goal:* Doctor records vitals, diagnoses, and prints/shares digital Rx in under 60 seconds.

- **User Stories & Engineering Tasks:**
    - `TASK-401`: Build Universal Vitals Grid (BP, Pulse, Weight, Height, BMI, Sugar).
    - `TASK-402`: Implement Specialist-specific Micro-Modules:
        - **Pediatrics:** Growth percentiles & IAP immunization checklist.
        - **Dermatology:** Anatomic body map & lesion morphology chip selectors.
        - **Orthopedics:** Joint locator, pain scale (VAS 1-10), and radiology orders.
        - **Diabetology:** Comorbidity pills & glycemic target history.
    - `TASK-403`: Build **Rapid Indian Rx Engine**:
        - Pre-loaded top 3,000 Indian pharmaceutical brands & generics.
        - Quick-tap dosage chips (`1-0-1`, `1-0-0`, `0-0-1`, `1-1-1`, `SOS`).
        - Meal timing chips (`After Food / Before Food`).
    - `TASK-404`: Auto-generate clean PDF prescription with doctor letterhead and WhatsApp dispatch.
- **Milestone Gate 4:** Doctor completes a full specialist consultation record and dispatches PDF Rx in under 60 seconds.

## Sprint 5: Payments, Thermal Printing & Ledger (Weeks 9–10)

*Primary Goal:* Complete financial reconciliation for cash and UPI.

- **User Stories & Engineering Tasks:**
  - `TASK-501`: Dynamic UPI QR Generator:
    - Dynamic BharatQR / UPI deep link based on exact consultation fee.
    - Customer-facing mobile screen or mini display QR rendering.
  - `TASK-502`: ESC/POS Thermal Printer Driver:
    - 2-inch and 3-inch thermal printer slips via Web Bluetooth / USB.
    - Prints clinic header, token number in large font, and tracking QR.
  - `TASK-503`: Daily Settlement Ledger:
    - One-click PDF/Excel export of daily OPD revenue (Cash vs. UPI split).
- **Milestone Gate 5:** Front-desk reconciles ₹15,000+ daily collection with zero cash mismatch.

### Sprint 6: Offline Resilience, UAT & Pilot Launch (Weeks 11–12)

*Primary Goal:* Deploy into a live operating specialist clinic for a 14-day test pilot.

- **User Stories & Engineering Tasks:**
  - `TASK-601`: Service Worker & IndexedDB offline cache:
    - If internet goes down, system queues all mutations locally.
    - Bi-directional background sync upon reconnection.
  - `TASK-602`: Pilot Clinic Deployment:
    - Install Smart TV app on clinic waiting room Android TV / FireStick.
    - Setup compounder PC and doctor iPad.
  - `TASK-603`: Live Shadow Testing:
    - Run ClinikFlow in parallel with paper register for 3 days.
    - Cut over to 100% digital operations.
- **Milestone Gate 6:** 500+ real patients processed through the system with zero downtime or lost records.

## 3. Definition of Done (DoD) Checklist

For every user story to be marked **Done**, it must satisfy:

- Mobile-tested on both iOS Safari and Android Chrome.
- Minimum touch target size $\geq 48\text{px} \times 48\text{px}$ ≥ 48px × 48px.
- Zero blocking network requests for local UI interactions.
- Real-time updates delivered to all connected screens in $< 300\text{ms}$ < 300ms.
- Error boundary implemented with user-friendly retry mechanism.
- Form validations respect Indian phone formats ( `+91` 10 digits).
- Tested under simulated 3G network conditions (latency 300ms, packet loss 2%).

## 4. Risk Matrix & Mitigation Strategies

| Risk Description | Severity | Probability | Mitigation Strategy |
|---|---|---|---|
| **Compounder resistance** (prefers paper register) | High | Medium | Involve compounder in UI design; enforce keyboard-only shortcuts ( `Alt+N` ); provide ₹500 weekly speed bonus during pilot. |
| **Clinic Wi-Fi downtime** | Critical | High | Local-first PWA architecture with IndexedDB; local LAN fallback via BroadcastChannel / local server. |
| **Doctor refuses to type** | High | High | Strict zero-typing rule. 95% of prescription and clinical notes are selected via pre-seeded chips and sliders. |
| **WhatsApp API delivery delays** | Medium | Medium | Fallback to transactional SMS (MSG91/Twilio) if WhatsApp webhook delivery receipt is not confirmed within 45s. |
| **TV Display browser sleeps / freezes** | Medium | Low | Kiosk mode wake-lock API ( `navigator.wakeLock.request('screen')` ) to prevent screen dimming. |